

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress
(256-bit S, K, T, r, C_mkt, TID)

Stage 1: Ingress
Registration (1 Cycle)

- AXI4-Stream 256-bit Ingress: {S, K, T, r, C_mkt, TID}
- Single-Cycle Register + TID Tag
 - No Ingress Divider
 - (Scale-Invariant Pad  Ratio eliminates 32-cyc divider)
- Q8.24 Fixed-Point Precision

Contract

Stage 2: Analytical Seeder
(Brenner-Subrahmanyam, 64 Cyc)

$$\sigma_0 = \frac{2.5066}{\sqrt{T}} \cdot \frac{C_{mkt}}{(S+K)/2}$$

- Shared Digit-Recurrence $\sqrt{T}$ (29 Cyc)
 - Forwards $\sqrt{T}$ Directly to Stage 3
- Internal Non-Restoring Divider (33 Cyc)
 - Bounds Init Error: $|\sigma_0 - \sigma^*| \leq 0.08$

$\sigma_0, \sqrt{T}$ (Fwd)

Stage 3: Core BS Datapath
(126 Cycles, II = 1)

- Scale-Invariant Pad  Ratio (S-K)/(S+K)
- 18-Cyc CORDIC Log Fallback (SRL15-Matched)
- Horner 5th-Order Poly CDF N(d1), N(d2)
- Hyperbolic CORDIC exp(-rT) & exp(-d1²/2)
- Fully Unrolled 126-Stage Feedforward

C_{BS}, ν
(126 Cyc)

Stage 4: NR Update & Dividers
(33 Cycles)

- Convergence: $|C_{BS} - C_{mkt}| \leq \0.01
- u_nr_divider: $\Delta\sigma = (C_{BS} - C_{mkt})/\nu$
- u_gamma_divider: $\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$
- Greeks: $\Delta = N(d_1)$, $\nu = S\sqrt{T}\phi(d_1)$
- TID-Scoreboard Priority Loopback

Converged (97.6%)

Stage 5: Multi-Core Arbiter
& 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Complete Drain
- Packed 128-Bit Single-Flit Bus:
 - [127:122] TID Transaction ID (6b)
 - [121:96] Gamma Greek (26b)
 - [95:64] Vega Greek (32b)
 - [63:32] Delta Greek (32b)
 - [31:0] sigma Implied Vol (32b)

AXI4-Stream Egress
(128-bit σ + Full Greeks)

Zero-BRAM Context Store

- 64 x 32-bit Distributed LUTRAM
- SRL32 Delay Shift Registers
- 0 Block RAMs (Zero-BRAM)
- Preserves 100% RAMB36/18 for 10GbE MAC & PCIe DMA

Context

Loopback
 σ_{n+1} (2.4%)